Raghuram Kalyanam

Senior Data Scientist/Consultant

German Citizen | Neudorfstr.10, 6313 Menzingen, CH | +(49) 797515004
raghuramkalyanam@gmail.com | linkedin.com/in/raghuramkalyanam | github.com/raghu1121

Senior Data Scientist / AI Engineer with 13 years of professional experience, including 10 years in machine learning and AI. Specialised in production LLM systems : RAG architecture and retrieval quality, evaluation frameworks and LLM-as-judge pipelines, prompt architecture, agentic workflows, and AI security and guardrails. Track record of taking GenAI systems from prototype to monitored production in regulated and public-sector environments, and of making architecture decisions on measured evidence rather than assumption. Earlier background in time-series forecasting, reinforcement learning and MLOps across finance, automotive, aviation and construction.

Work Experience

October 2025 – Present

Senior Data Scientist / Consultant, ELCA Informatique SA, Zurich

- Architected and delivered three production GenAI systems : a sovereign, fully self-hosted assistant platform for the **Canton of St. Gallen** handling classified data, plus an internal staff assistant and a public-facing multilingual chatbot for a **leading Swiss collection agency**.
- Cut cost per request 26× (64.5k to 2.4k tokens) while raising rule compliance from 60% to nearly 100% and latency by 17%, after root-causing non-deterministic chunk retrieval that made mandatory rules intermittently invisible to the model; proven with a controlled 24-run experiment that also disproved the tool-calling alternative.
- Re-architected retrieval from a two-pass LLM-as-navigator design to BM25 over OpenSearch with per-language analysers : mean latency 97s to 11–16s, tokens per question down 70%, and retrieval hit rate 0.70 to 0.95 via German compound-aware query shaping and LLM-generated section tags.
- Built evaluation frameworks from scratch (61 scenarios, 344 weighted assertions, cross-family LLM judge, MLflow per-question tracking, retrieval-vs-generation failure attribution) and fixed a harness that had been silently measuring a straw-man retrieval path.
- Designed multimodal retrieval without a vision model : 2,110 screenshots across 498 sections attached deterministically from an index, making hallucinated image references structurally impossible.
- Owned security for the public-facing chatbot : nine-stage defence pipeline against prompt injection and abuse, 177-test suite driven by a 37-case attack corpus, and 10 of 15 external penetration-test findings closed with per-finding regression tests; mapped OWASP LLM Top-10 and 21 data-protection measures onto implemented controls.
- Productionised the platform layer : zero-downtime reindexing via atomic index aliasing, Keycloak OIDC SSO with directory-group RBAC, Prometheus/Grafana observability with per-use-case cost attribution, and GitLab CI/CD. A deployment audit caught defects destroying chat history on every release and exposing the search cluster without TLS.
- Rejected two plausible architectures on measured evidence – cross-encoder reranking (deployed, evaluated, net-neutral) and vector search (lexical retrieval measurably better on a curated German corpus) – and documented both decisions.

LLMs | RAG | OpenSearch | Open WebUI | LiteLLM | Bifrost | Ollama | vLLM | Qwen3 | Gemma | DeepEval | MLflow | Chainlit | FastAPI | Keycloak | Prometheus | Grafana | Docker | GitLab CI/CD | Azure OpenAI | Python | Prompt Engineering | LLM-as-Judge | Guardrails

July 2022 – September 2025

Senior Data Scientist, Mercedes-Benz Mobility AG, Stuttgart

- Built an LLM-powered semantic query system letting collections staff identify high-risk customers in natural language, using Model Context Protocols to translate intent into weighted SQL and vector queries across risk history, payment profiles and behavioural data.
- Built a semantic fraud reasoning system with hybrid search over analyst comments and structured financial metrics; used FAISS to cluster historical fraud patterns and LangGraph to orchestrate multi-step, traceable risk assessments.
- Integrated Whisper to transcribe and semantically index customer conversations, making them searchable by tone, intent and risk.

LLMs | LangChain | LangGraph | MCPs | ChromaDB | pgvector | FAISS | Superlinked | Whisper | Ollama | LiteLLM | OpenAI API | Hugging Face | FastAPI | Docker | PySpark | SQL

January 2020 – May 2022 | **Data Scientist / Consultant, Lufthansa Industry Solutions GmbH, Raunheim**

- Developed an AI forecasting platform with rolling predictions, automated retraining and Bayesian hyperparameter optimisation, plus the ETL workflows enriching data with external regressors.
- Modelled resource and scheduling problems as reinforcement learning tasks solved with DQN variants; containerised training and inference on Docker and Kubernetes for production.
- Consulted aviation, automotive and public-sector clients; mentored junior data scientists and advised on ML architecture.

Python | PyTorch | TensorFlow | scikit-learn | Prophet | skforecast | MLflow | FastAPI | Docker | Kubernetes | Azure | Terraform | PySpark | SQL

May 2015 – October 2019 | **Research Associate, TU Kaiserslautern | LivingLab SmartOfficeSpace, Kaiserslautern**

- Developed reinforcement learning models controlling adaptive glass transparency to optimise visual and thermal comfort in real office environments, with a purpose-built daylight simulation environment generating synthetic training data and large-scale tuning via Ray Tune.
- Collaborated with SBRC (University of Wollongong, Australia) on a co-simulation framework for building performance and user comfort; six peer-reviewed publications.

Python | PyTorch | TensorFlow | Keras | RLlib | Ray Tune | OpenAI Gym | Flask | Spark | Hadoop | SQLite

November 2012 – February 2015 | **Software Engineer, Ericsson R&D Center, Chennai, India**

- Implemented VoLTE (Java) and Diameter (C++) protocols for telecom billing systems in an agile team; node performance improvements, on-call support for international customers, and release management.

Java | C++ | Design Patterns | Postgres | Wireshark | Jenkins | Jira | Git

</> Skills

Programming Languages	Python, SQL, Java, C++.
GenAI / LLM	LangChain, LangGraph, MCPs, OpenAI API, Azure OpenAI, LiteLLM, Bifrost, Ollama, vLLM, Open WebUI, Chainlit, Qwen3, Gemma, Mistral, LLaMA, Hugging Face Transformers.
RAG & Retrieval	RAG architecture, OpenSearch, BM25 & lexical analysers, hybrid search, query expansion & rewriting, reranking, chunking strategies, multimodal retrieval, ChromaDB, pgvector, FAISS, Superlinked, Docling, Scrapy, BeautifulSoup.
Evaluation & Quality	DeepEval, LLM-as-judge, GEval, golden datasets, deterministic rule metrics, MLflow experiment tracking, retrieval failure attribution, A/B experiments, prompt architecture.
AI Security	Prompt-injection guardrails, OWASP Top-10 for LLM Applications, adversarial test corpora, groundedness & hallucination controls, Keycloak OIDC, RBAC, DPIA-driven controls.
ML & Optimisation	PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, CatBoost, RLlib, Prophet, skforecast, Ray Tune, HyperOpt, Bayesian optimisation.
Cloud & MLOps	Azure, Docker, Docker Compose, Kubernetes, Helm, Terraform, GitLab CI/CD, Prometheus, Grafana, FastAPI, Flask, Git.
Data & Visualisation	PySpark, Pandas, NumPy, Power BI, Tableau, Matplotlib, Seaborn, Plotly, Streamlit, Whisper.

Education

2007 - 2012 **M.Sc Mathematics and Computing**, IIT Dhanbad, India.

Achievements

- Six peer-reviewed publications in reinforcement learning and building-performance modelling (see below).
- Ranked in the top 1% of 400,000 candidates in the IIT-JEE entrance examination.
- Held a merit scholarship throughout my studies; recognised as “Rockstar” at Ericsson for performance.

Languages

English	Fluent
German	C1

Publications

- Kalyanam, Raghuram, and Sabine Hoffmann. 2021. ”A Reinforcement Learning- Based Approach to Automate the Electrochromic Glass and to Enhance the Visual Comfort” *Applied Sciences* 11, no. 15 : 6949.
- Kalyanam, Raghuram, and Sabine Hoffmann. ”Reinforcement learning based approach to automate the external shading system and enhance the occupant comfort.” *AI in AEC conference*, Helsinki, Finland, March 2021.
- Kalyanam, Raghuram, and Sabine Hoffmann. ”A Novel Approach to Enhance the Generalization Capability of the Hourly Solar Diffuse Horizontal Irradiance Models on Diverse Climates.” *Energies* 13.18 (2020) : 4868.
- Kalyanam, Raghuram and Sabine Hoffmann. 2016. ”Visual and thermal comfort with electrochromic glass using an adaptive control strategy.” Paper presented at the *15th International Radiance Workshop (IRW)*, Padua, Italy, August 2016.
- Boudier, Katharina, Massimo Fiorentini, Sabine Hoffmann, Raghuram Kalyanam, and Georgios Kokogiannakis. 2016. ”Coupling a thermal comfort model with building simulation for user comfort and energy efficiency.” *In Proceedings of the Central European Symposium on Building Physics (CESBP) and BauSIM*, Dresden, Germany, September 2016, 481–487.
- Hoffmann, Sabine, Andrew McNeil, Eleanor S. Lee, and Raghuram Kalyanam. 2015. ”Discomfort glare with complex fenestration systems and the impact on energy use when using daylighting control.” *In Proceedings of the Advanced Building Skins Conference*, Bern, Switzerland, November 2015.